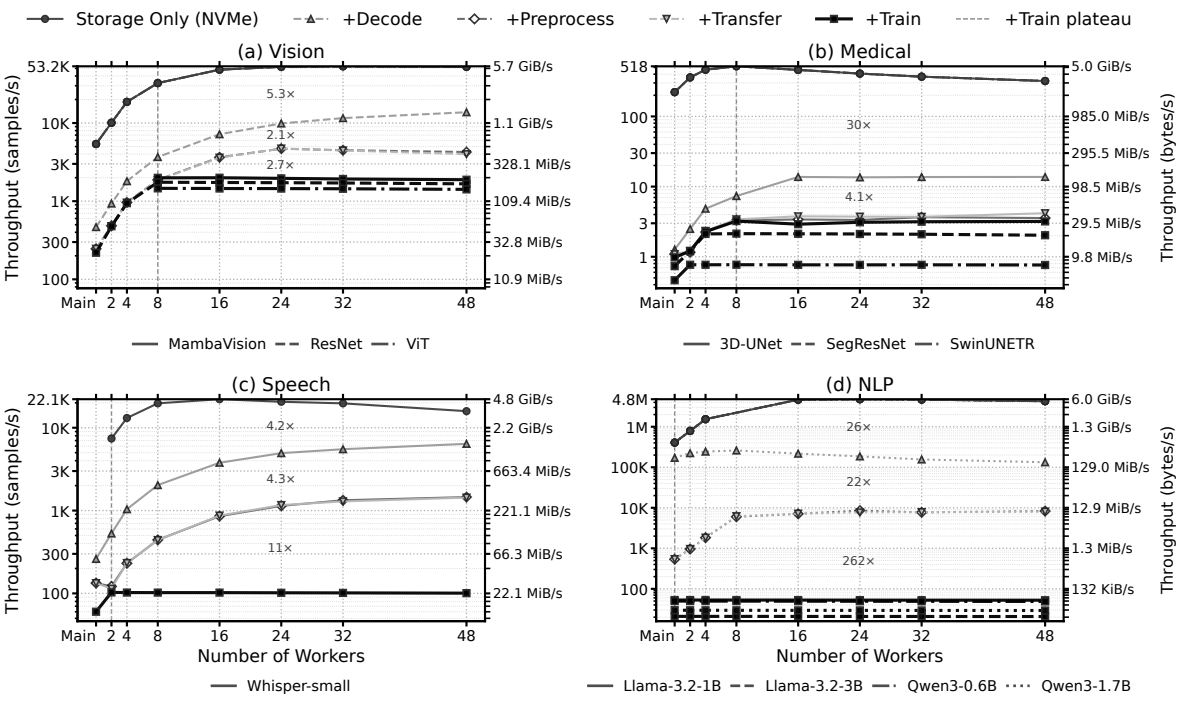
Storage Only (NVMe)
+Decode
+Preprocess
+Transfer
+Train
+Train plateau
(a) Vision
Throughput (samples/s)
53.2K
10K
3K
1K
300
100
5.7 GiB/s
1.1 GiB/s
328.1 MiB/s
109.4 MiB/s
32.8 MiB/s
10.9 MiB/s
5.3×
2.1×
2.7×
Main 2 4 8 16 24 32 48
MambaVision
ResNet
ViT
(b) Medical
Throughput (bytes/s)
518
100
30
10
3
1
5.0 GiB/s
985.0 MiB/s
295.5 MiB/s
98.5 MiB/s
29.5 MiB/s
9.8 MiB/s
30×
4.1×
3D-UNet
SegResNet
SwinUNETR
(c) Speech
Throughput (samples/s)
22.1K
10K
3K
1K
300
100
4.8 GiB/s
2.2 GiB/s
663.4 MiB/s
221.1 MiB/s
66.3 MiB/s
22.1 MiB/s
4.2×
4.3×
11×
Number of Workers
Whisper-small
(d) NLP
Throughput (bytes/s)
4.8M
1M
100K
10K
1K
100
6.0 GiB/s
1.3 GiB/s
129.0 MiB/s
12.9 MiB/s
1.3 MiB/s
132 KiB/s
26×
22×
262×
Number of Workers
Llama-3.2-1B
Llama-3.2-3B
Qwen3-0.6B
Qwen3-1.7B